Stony Brook Institute at Anhui University

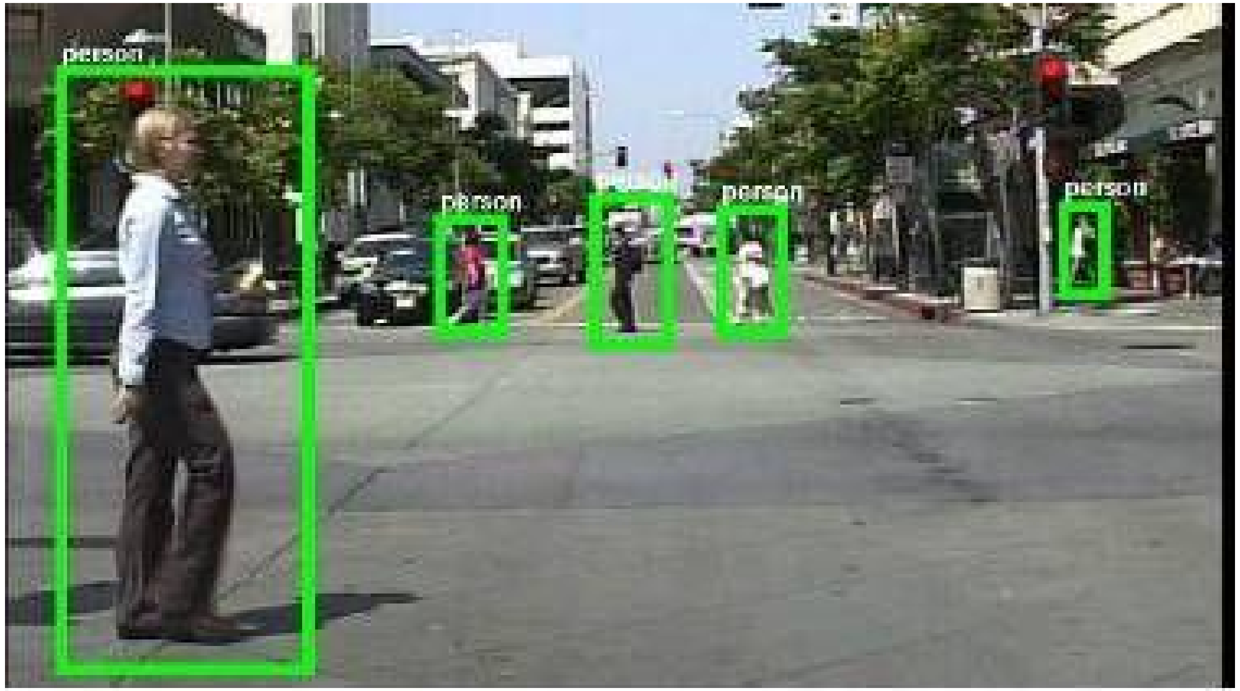
安徽大学纽约石溪学院

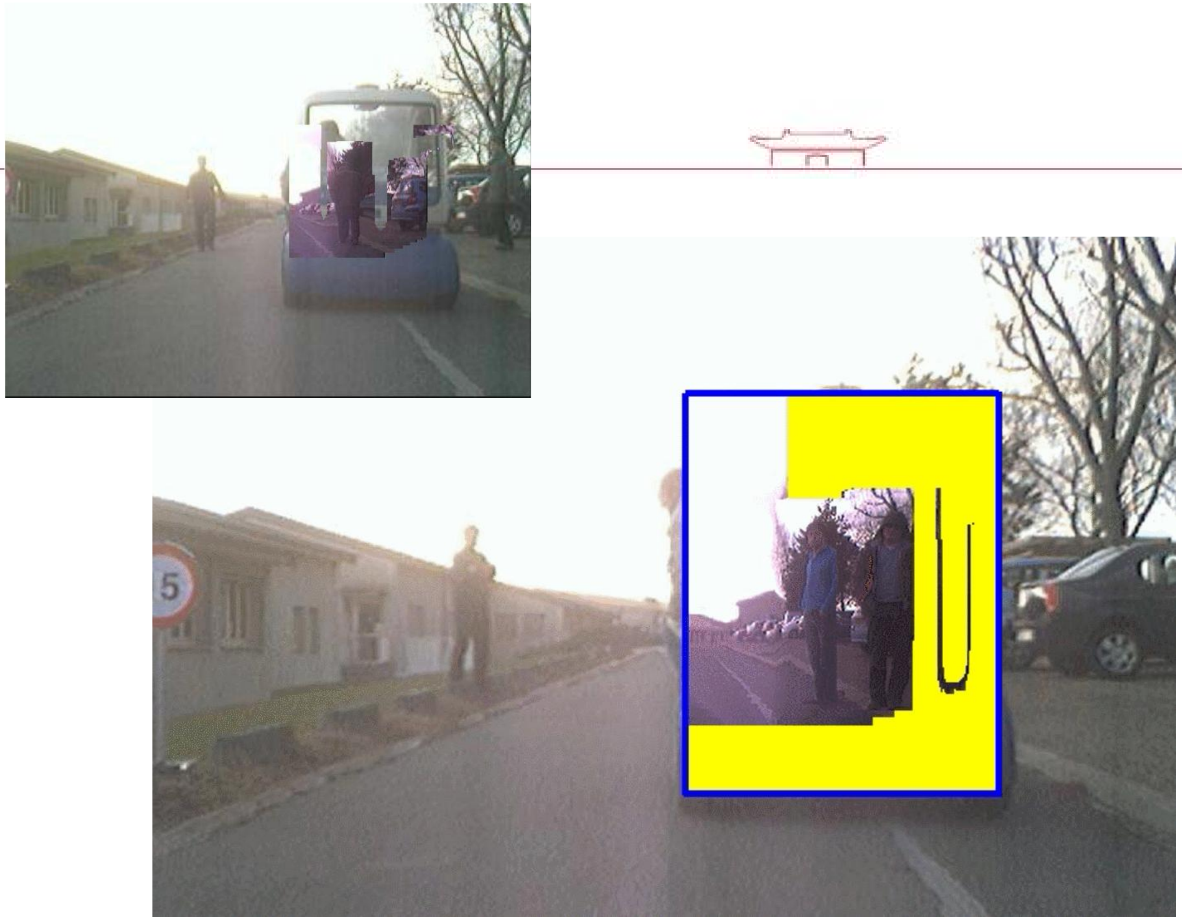
# ISE 333: User Interface Development (Design) Practice 2: UI-Oriented Image Processing

Spring 2026

Instructor: Hao Li

# Before the journey …

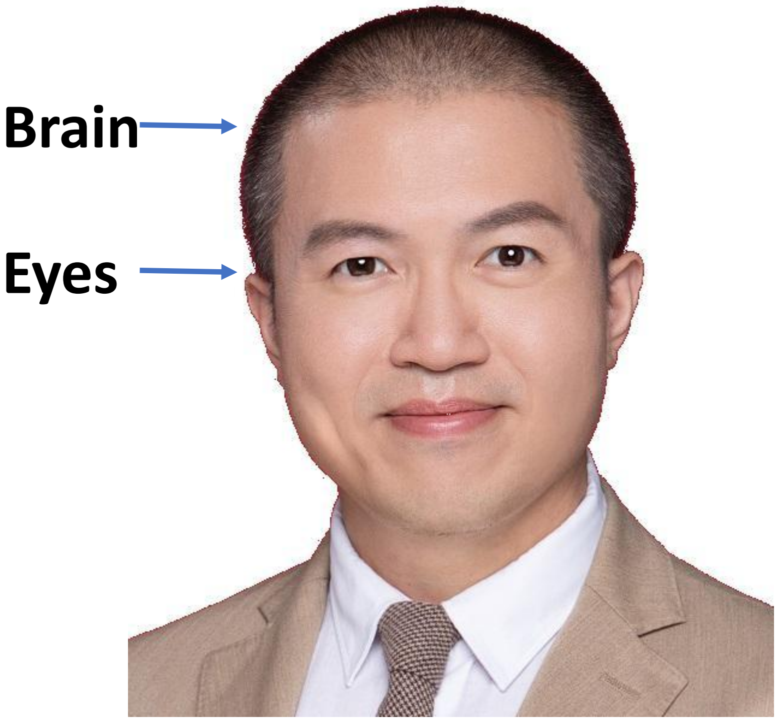


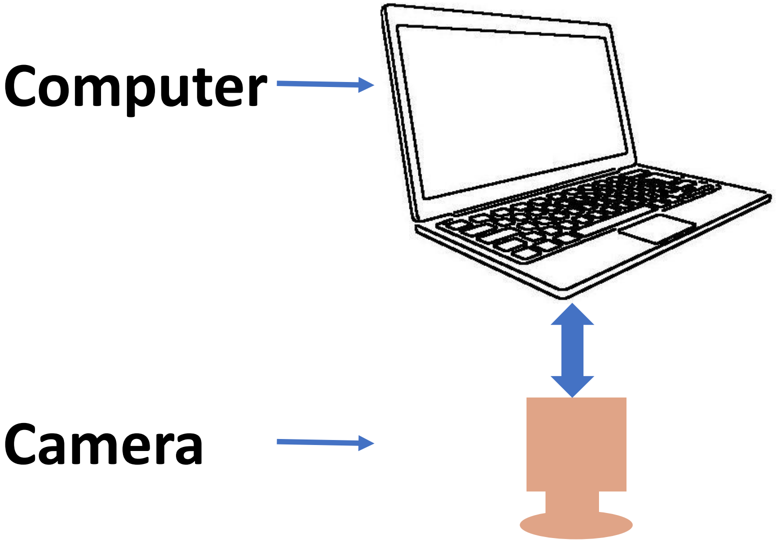

How to add the AR bounding boxes?

How to generate the see through effect?

# Vision Systems

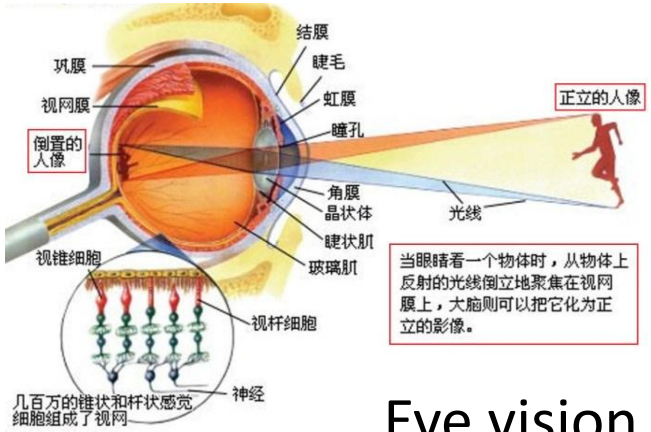



biological vision system

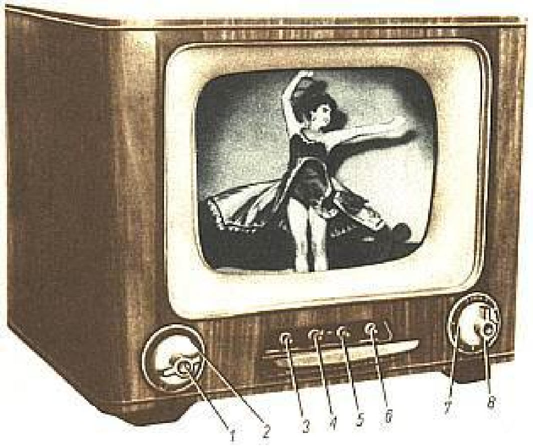



computer vision system

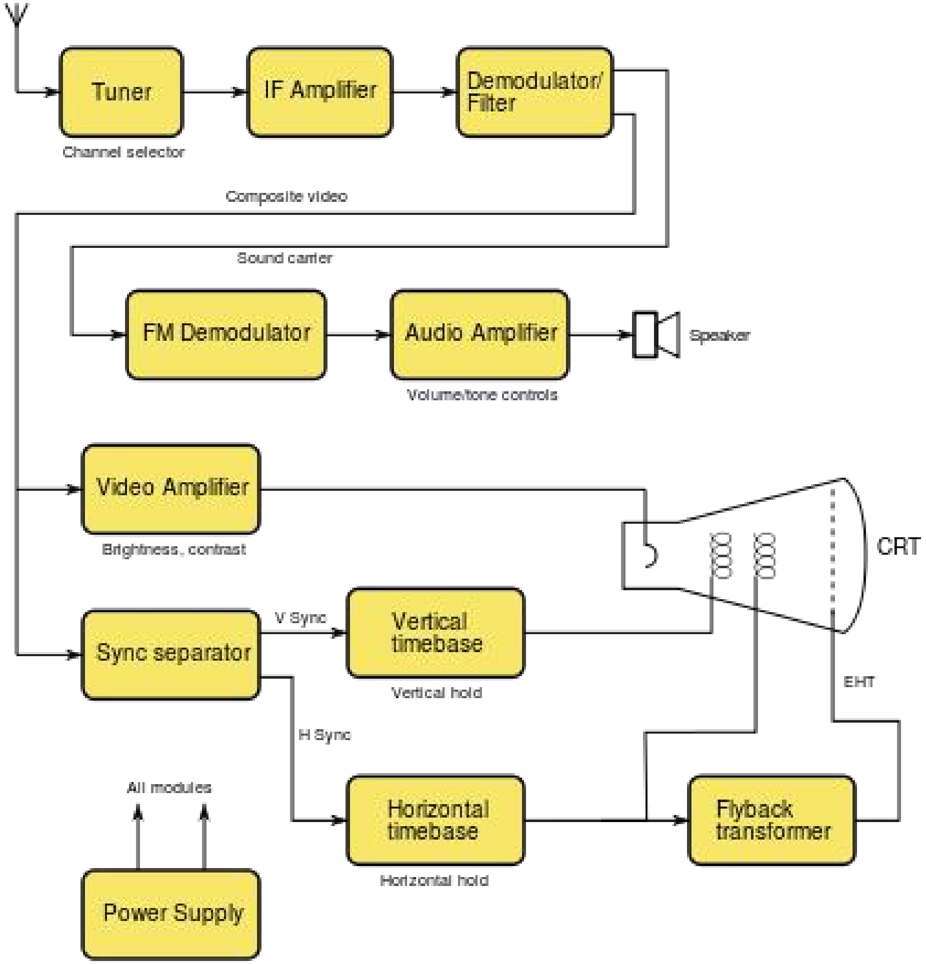
# What Is Image Processing?



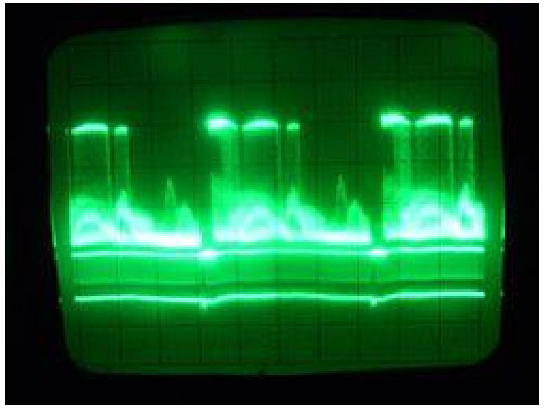

Eye vision - biological



Analog TeleVision



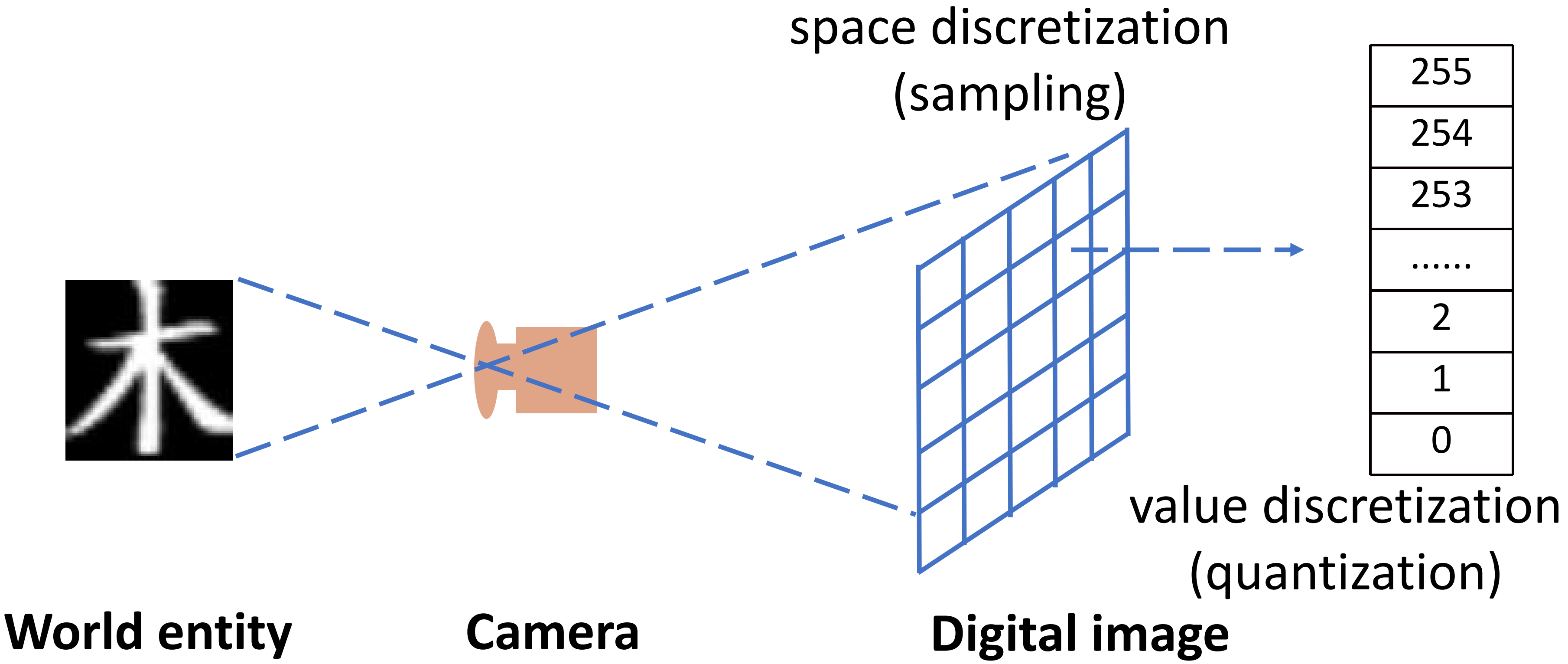

Washing in photography - chemical



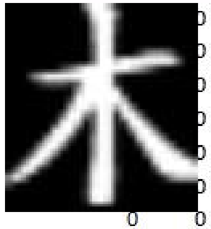
**Analog image signal processing**

# What Is Image Processing?

Analog Image ⟶ Discretization ⟶ Digital Image



# What Is Image Processing?



| | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 141 | 255 | 255 | 192 | 43 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 34 | 238 | 255 | 255 | 128 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 4 | 178 | 255 | 248 | 54 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 141 | 255 | 248 | 54 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 141 | 255 | 243 | 22 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 98 | 255 | 243 | 22 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 98 | 255 | 243 | 22 | 0 | 4 | 62 | 140 | 153 | 153 | 107 | 11 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 98 | 255 | 255 | 197 | 255 | 255 | 255 | 255 | 255 | 255 | 248 | 115 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 4 | 62 | 116 | 184 | 255 | 255 | 255 | 255 | 255 | 255 | 214 | 166 | 107 | 47 | 11 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 104 | | | | | | | | | | | | | | | | | | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | | | | | | | | | | | | | | | | | | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | | | | | | | | | | | | | | | | | | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | | | | | | | | | | | | | | | | | | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 4 | 178 | 255 | 248 | 204 | 255 | 243 | 165 | 255 | 128 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 98 | 255 | 255 | 128 | 98 | 255 | 243 | 30 | 225 | 255 | 128 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 34 | 238 | 255 | 205 | 11 | 98 | 255 | 243 | 22 | 60 | 255 | 255 | 128 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 8 | 225 | 255 | 248 | 54 | 0 | 98 | 255 | 243 | 22 | 0 | 141 | 255 | 255 | 128 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 4 | 178 | 255 | 255 | 89 | 0 | 0 | 141 | 255 | 243 | 22 | 0 | 8 | 225 | 255 | 255 | 128 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 141 | 255 | 255 | 128 | 0 | 0 | 0 | 141 | 255 | 243 | 22 | 0 | 0 | 60 | 255 | 255 | 255 | 157 | 11 | 0 | 0 |
| 0 | 0 | 0 | 98 | 255 | 255 | 128 | 0 | 0 | 0 | 0 | 141 | 255 | 243 | 22 | 0 | 0 | 0 | 141 | 255 | 255 | 255 | 209 | 54 | 0 |
| 0 | 0 | 98 | 255 | 255 | 128 | 0 | 0 | 0 | 0 | 4 | 178 | 255 | 243 | 22 | 0 | 0 | 0 | 8 | 225 | 255 | 255 | 255 | 255 | 157 |
| 0 | 98 | 255 | 248 | 54 | 0 | 0 | 0 | 0 | 0 | 8 | 225 | 255 | 248 | 54 | 0 | 0 | 0 | 0 | 60 | 255 | 255 | | | |
| 134 | 248 | 157 | 11 | 0 | 0 | 0 | 0 | 0 | 0 | 34 | 238 | 255 | 248 | 54 | 0 | 0 | 0 | 0 | 0 | 40 | 140 | | | |
| 69 | 11 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 34 | 238 | 255 | 243 | 22 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | | | |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 60 | 255 | 255 | 243 | 22 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | | | |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 34 | 238 | 255 | 205 | 11 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | | | |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | | | |

```
img = imread('character_mu.png');
img_reverse = 255-img; imshow(img_reverse);
```

**Manipulation of the digital matrix to achieve certain objective**

# Color Image Processing

Color I(:, :, [R, G, B])

```
img = imread('Suisse.jpg');
disp(size(img));
imgR = uint8(zeros(size(img)));
imgG = imgR; imgB = imgR;
imgR(:, :, 1) = img(:, :, 1);
imgR(:, :, [2, 3]) = 0; % Red channel
imgG(:, :, 2) = img(:, :, 2);
imgG(:, :, [1, 3]) = 0; % Green channel
imgB(:, :, 3) = img(:, :, 3);
imgB(:, :, [1, 2]) = 0; % Blue channel
figure(1), subplot(2, 2, 1), imshow(img);
subplot(2, 2, 2), imshow(imgR);
subplot(2, 2, 3), imshow(imgG);
subplot(2, 2, 4), imshow(imgB);
```

# Color-To-Gray Conversion

Color I(:, :, [R, G, B]) → Gray I(:, :)

G = 0.2989 * R + 0.5870 * G + 0.1141 * B

```
img = imread('Suisse.jpg');
imgD = double(img);
imgD2 = 0.2989*imgD(:,:,1) + 0.5870*imgD(:,:,2) + 0.1141*imgD(:,:,3);
img2 = uint8(imgD2);
figure(1), subplot(2,1,1), imshow(img);
subplot(2,1,2), imshow(img2);

img3 = rgb2gray(img); figure(2), imshow(img3);
```

# Thresholding

Given a digital image represented as a matrix I(:,:), select a threshold T; for an arbitrary pixel (r,c):
if I(r,c) >= T, set I(r,c) = 1 (or 255);
otherwise, set I(r,c) = 0.

## Image Binarization

| | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 141 | 255 | 255 | 192 | 43 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 34 | 238 | 255 | 255 | 128 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 4 | 178 | 255 | 248 | 54 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 141 | 255 | 248 | 54 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 141 | 255 | 243 | 22 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 98 | 255 | 243 | 22 | 0 | 0 | 0 | 0 | | | | | | |
| | | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 98 | 255 | 243 | 22 | 0 | 4 | 62 | 140 | | | | | | |
| | | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 98 | 255 | 255 | 197 | 255 | 255 | 255 | 255 | | | | | | |
| | | 0 | 0 | 0 | 0 | 4 | 62 | 116 | 184 | 255 | 255 | 255 | 255 | 255 | 255 | 214 | 166 | 107 | | | | | | |
| 0 | 0 | 0 | 104 | 184 | 255 | 255 | 255 | 255 | 255 | 255 | 255 | 255 | 243 | 43 | 0 | 0 | 0 | 0 | | | | | | |
| 0 | 0 | 0 | 0 | 40 | 73 | 73 | 73 | 47 | 98 | 255 | 255 | 255 | 248 | 54 | 0 | 0 | 0 | 0 | | | | | | |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 4 | 178 | 255 | 255 | 255 | 255 | 243 | 22 | 0 | 0 | 0 | | | | | | |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 60 | 255 | 255 | 255 | 255 | 255 | 255 | 174 | 0 | 0 | 0 | | | | | | |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 4 | 178 | 255 | 248 | 204 | 255 | 243 | 165 | 255 | 128 | 0 | 0 | | | | | | |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 98 | 255 | 255 | 128 | 98 | 255 | 243 | 30 | 225 | 255 | 128 | 0 | | | | | | |
| 0 | 0 | 0 | 0 | 0 | 0 | 34 | 238 | 255 | 205 | 11 | 98 | 255 | 243 | 22 | 60 | 255 | 255 | 128 | | | | | | |
| 0 | 0 | 0 | 0 | 0 | 8 | 225 | 255 | 248 | 54 | 0 | 98 | 255 | 243 | 22 | 0 | 141 | 255 | 255 | | | | | | |
| 0 | 0 | 0 | 0 | 4 | 178 | 255 | 255 | 89 | 0 | 0 | 141 | 255 | 243 | 22 | 0 | 8 | 225 | 255 | | | | | | |
| 0 | 0 | 0 | 0 | 141 | 255 | 255 | 128 | 0 | 0 | 0 | 141 | 255 | 243 | 22 | 0 | 0 | 60 | 255 | | | | | | |
| 0 | 0 | 0 | 98 | 255 | 255 | 128 | 0 | 0 | 0 | 0 | 141 | 255 | 243 | 22 | 0 | 0 | 0 | 141 | | | | | | |
| 0 | 0 | 98 | 255 | 255 | 128 | 0 | 0 | 0 | 0 | 4 | 178 | 255 | 243 | 22 | 0 | 0 | 0 | 8 | | | | | | |
| 0 | 98 | 255 | 248 | 54 | 0 | 0 | 0 | 0 | 0 | 8 | 225 | 255 | 248 | 54 | 0 | 0 | 0 | 0 | | | | | | |
| 134 | 248 | 157 | 11 | 0 | 0 | 0 | 0 | 0 | 0 | 34 | 238 | 255 | 248 | 54 | 0 | 0 | 0 | 0 | | | | | | |
| 69 | 11 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 34 | 238 | 255 | 243 | 22 | 0 | 0 | 0 | 0 | | | | | | |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 60 | 255 | 255 | 243 | 22 | 0 | 0 | 0 | 0 | | | | | | |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 34 | 238 | 255 | 205 | 11 | 0 | 0 | 0 | 0 | | | | | | |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | | | | | | |

| | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 1 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 |
| 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 0 |
| 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 |
| 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

```
if I(r,c)>= 220
        I(r,c) = 1;
else    I(r,c) = 0
```

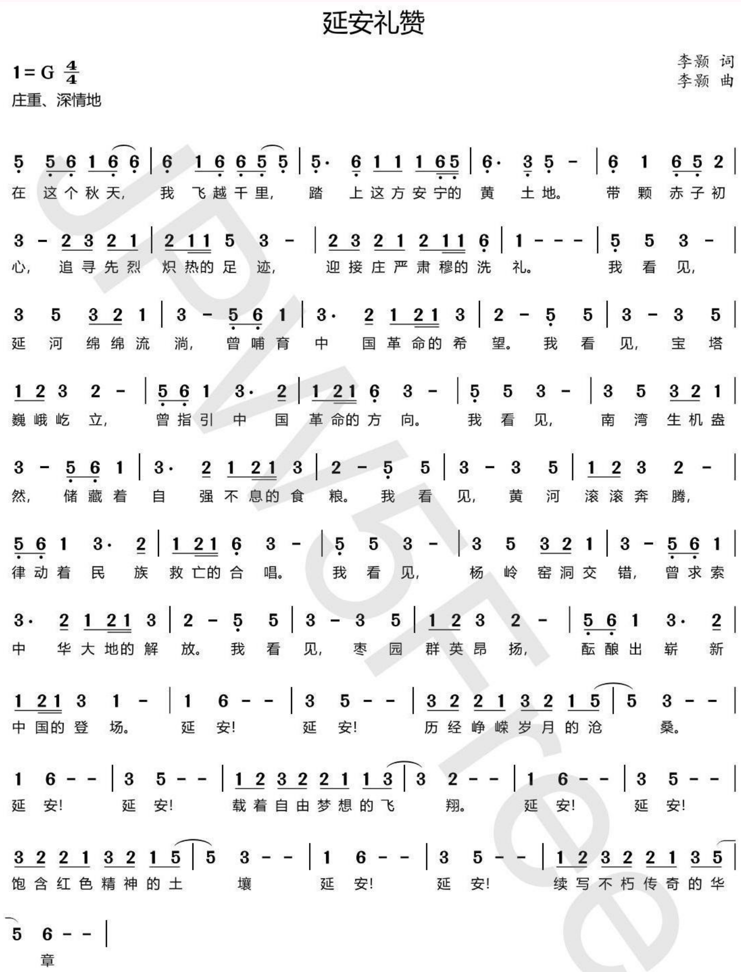
# Thresholding

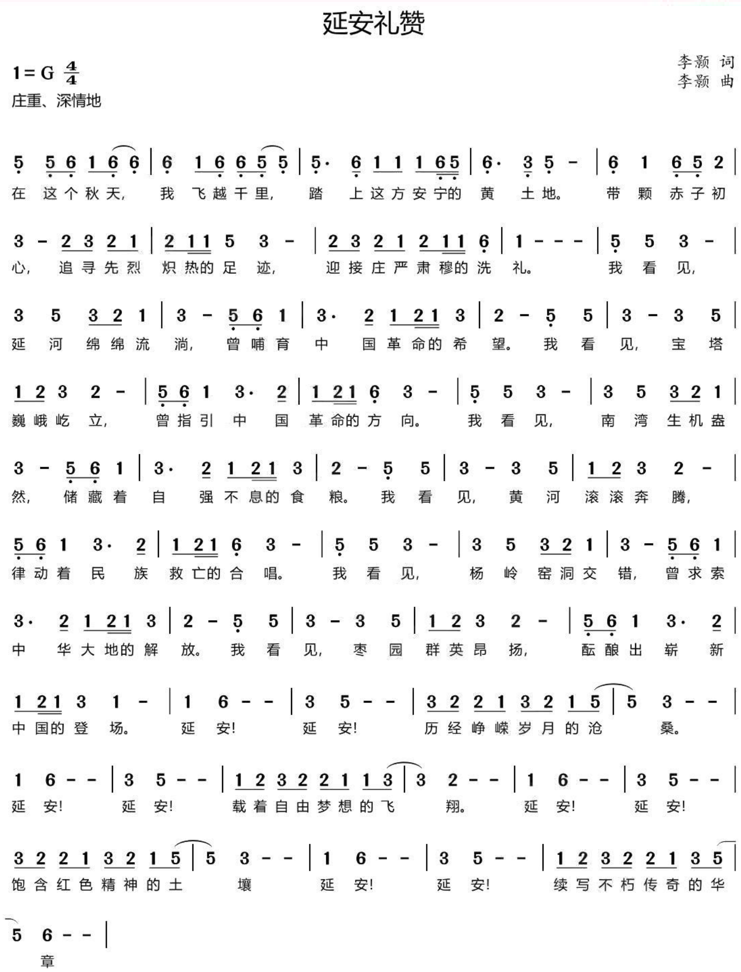
Given a digital image represented as a matrix I(:,:), select a threshold T; for an arbitrary pixel (r,c):
if I(r,c) >= T, set I(r,c) = 1 (or 255);
otherwise, set I(r,c) = 0.

**How to select the threshold?**

- Select a threshold empirically
  e.g. medical analysis, active photography
- Adaptive thresholding
  e.g. T = mean(I(:,:))

# Practice: Remove the water proof




Download *RmWaterproof_practice.m* and complete it

DISCUSSION

Q & A

# Topological Processing

## Pixel-wise processing

The image is processed pixel by pixel.
Each pixel is processed individually or independently.

## Topological processing

Processing of each pixel involves the relationship between the pixel and its **neighborhood** i.e. **neighbor pixels**.

# Non-Maximum Suppression

$$\mathbf{I}(x, y) = 0$$
$$\text{if } \mathbf{I}(x, y) < \max\{\mathbf{I}(x-1:x+1, y-1:y+1)\}$$

Obtain extrema

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 2 |
| 1 | 2 | 2 | 1 | 1 | 1 | 1 | 2 | 3 |
| 1 | 1 | 1 | 1 | 2 | 2 | 1 | 1 | 2 |
| 1 | 1 | 1 | 2 | 3 | 3 | 2 | 1 | 1 |
| 1 | 1 | 2 | 3 | 4 | 4 | 3 | 2 | 1 |
| 1 | 2 | 3 | 4 | 5 | 4 | 3 | 2 | 1 |
| 1 | 1 | 2 | 3 | 4 | 3 | 2 | 1 | 2 |
| 1 | 2 | 1 | 2 | 3 | 2 | 1 | 2 | 3 |
| 1 | 1 | 1 | 1 | 2 | 1 | 1 | 1 | 2 |

NMS →

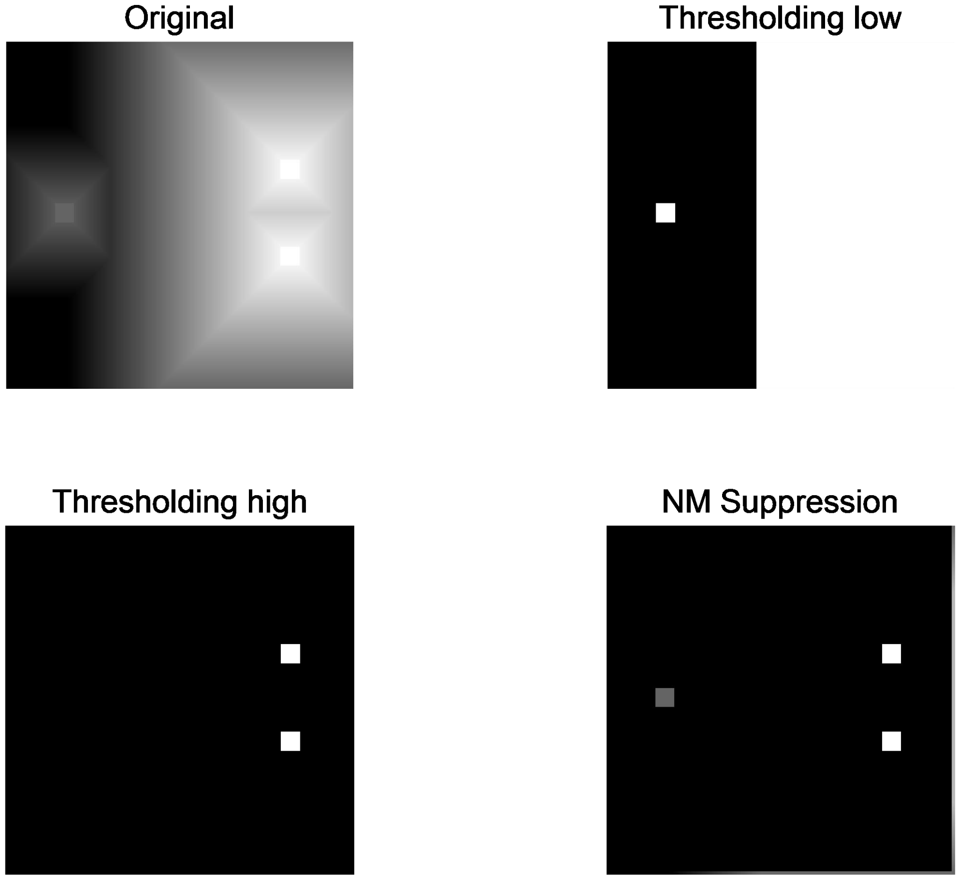
| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 2 | 2 | 0 | 0 | 0 | 0 | 0 | 3 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 5 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 2 | 0 | 0 | 0 | 0 | 0 | 0 | 3 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

# Non-Maximum Suppression

$$\mathbf{I}(x,y)=0$$

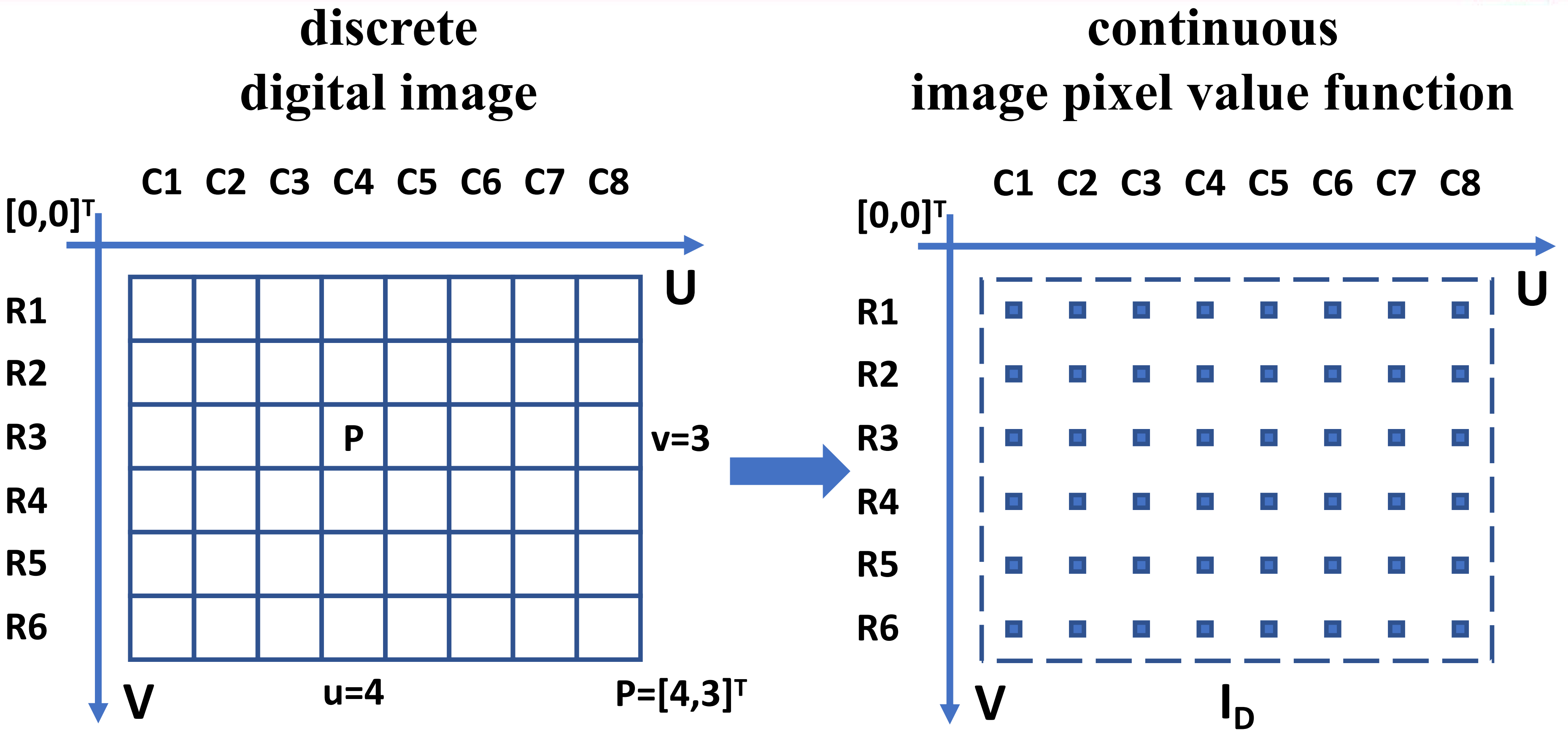
$$\text{if } \ \mathbf{I}(x,y)<\max\{\mathbf{I}(x-1:x+1,y-1:y+1)\}$$

Obtain extrema




Download *NonMaximumSuppression_practice.m* and complete it

# Image Coordinates System




$\bar{I}(u,v)$

$I(u,v) = \bar{I}(u,v)$ For each $[u,v]^T \in \mathbf{I}_D$

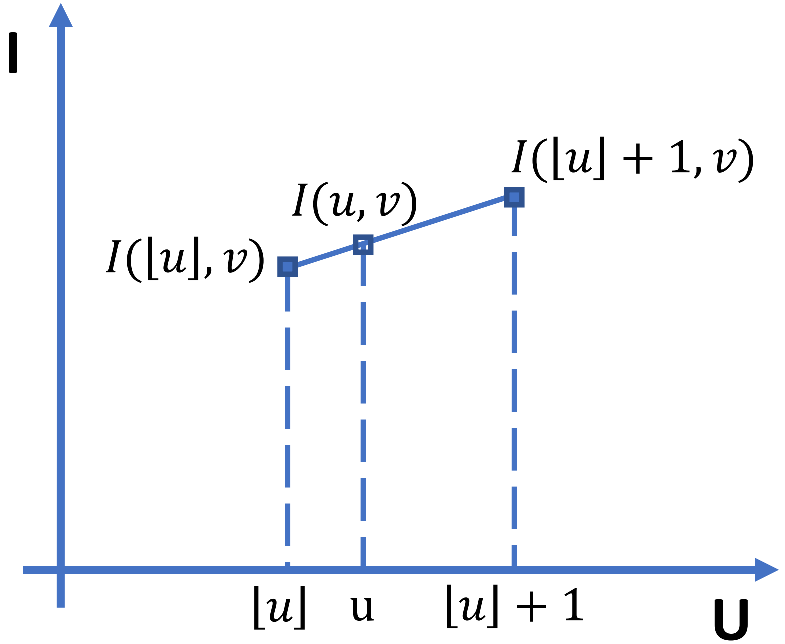
$\mathbf{I}_D = \{[u,v]^T | u \in \{1,2,\cdots,W\}, v \in \{1,2,\cdots,H\}\}$

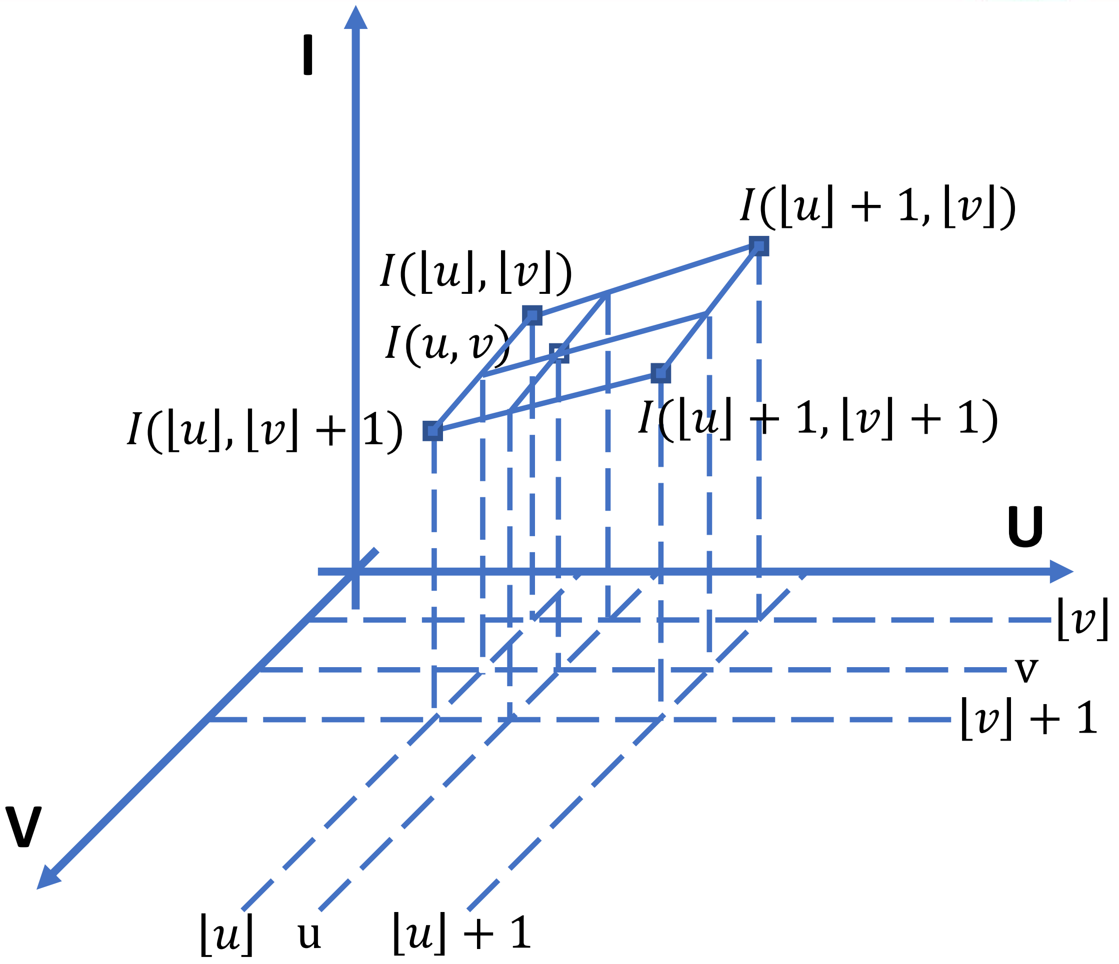
$\mathbf{I}_C = \{[u,v]^T | 1 \leq u \leq W, 1 \leq v \leq H\}$

# Image Coordinates System

**Linear Interpolation**

- one-dimensional
- two-dimensional



# Image Coordinates System

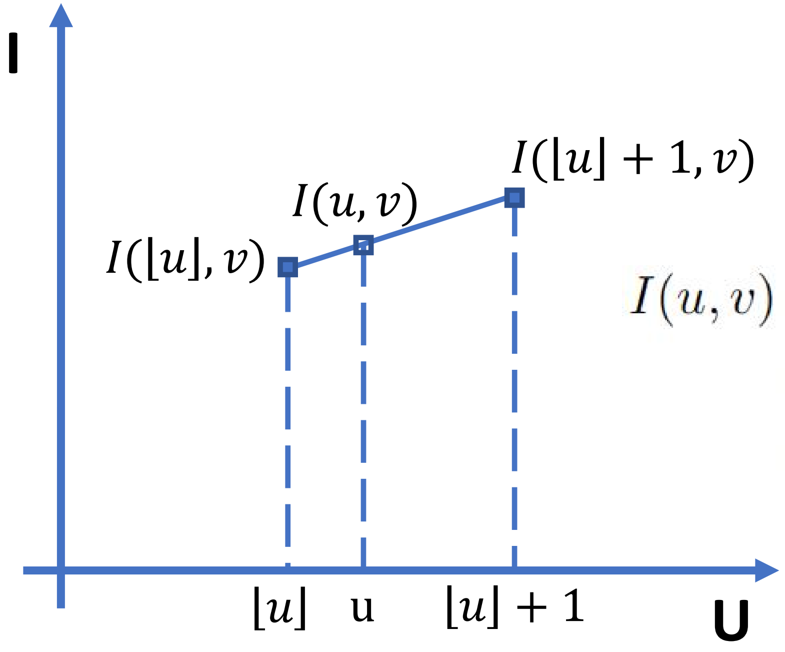
**Linear Interpolation**
- one-dimensional
- two-dimensional

$$I(u,v) = a_1 u + a_0$$

$$\Updownarrow$$

$$I(u,v) = (\lfloor u \rfloor + 1 - u)k_1 + (u - \lfloor u \rfloor)k_2$$

$$\begin{cases} (\lfloor u \rfloor + 1 - u)k_1 + (u - \lfloor u \rfloor)k_2|_{u=\lfloor u \rfloor} & = I(\lfloor u \rfloor, v) \\ (\lfloor u \rfloor + 1 - u)k_1 + (u - \lfloor u \rfloor)k_2|_{u=\lfloor u \rfloor + 1} & = I(\lfloor u \rfloor + 1, v) \end{cases}$$



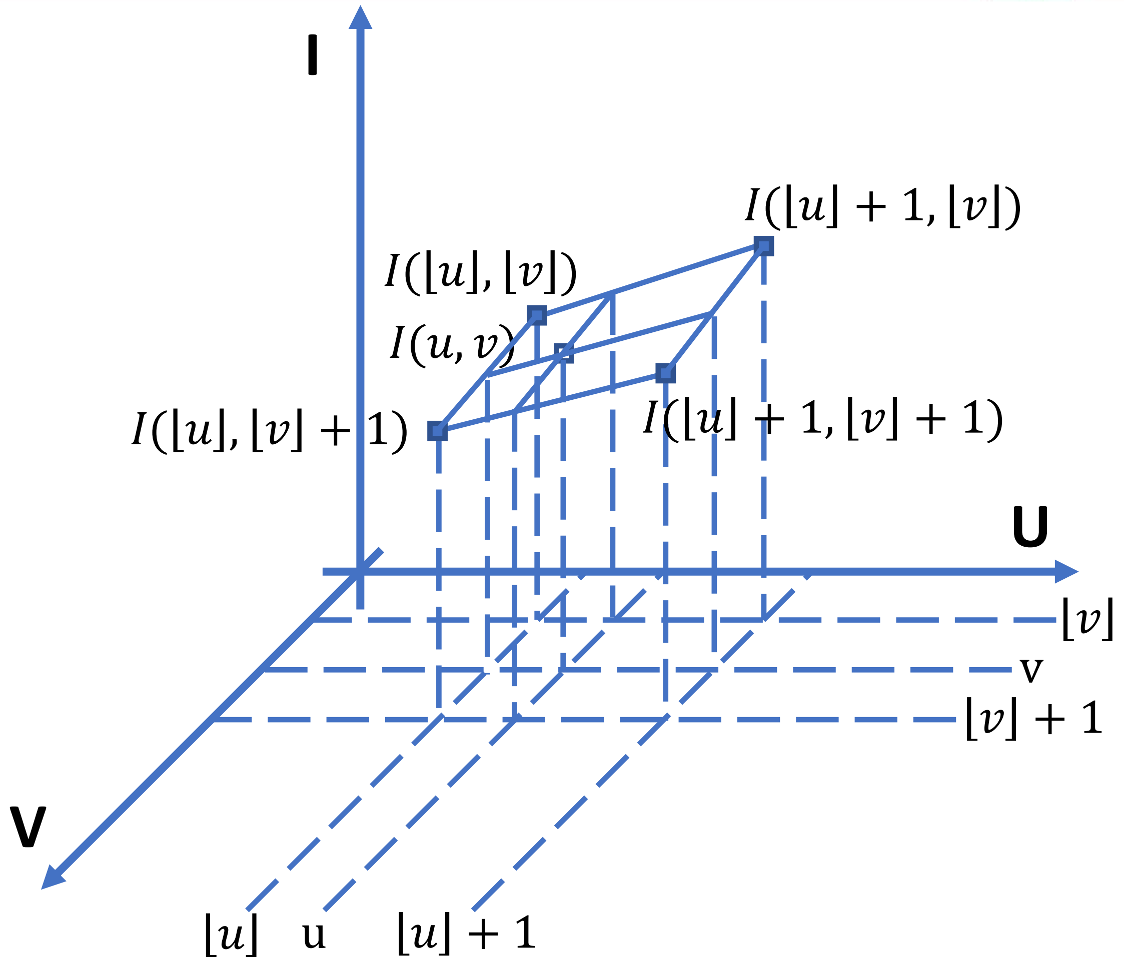

$$\begin{aligned} I(u,v) &= (\lfloor u \rfloor + 1 - u)k_1 + (u - \lfloor u \rfloor)k_2 \\ &= (\lfloor u \rfloor + 1 - u)I(\lfloor u \rfloor, v) + (u - \lfloor u \rfloor)I(\lfloor u \rfloor + 1, v) \\ &= (\lfloor u \rfloor + 1 - u)\bar{I}(\lfloor u \rfloor, v) + (u - \lfloor u \rfloor)\bar{I}(\lfloor u \rfloor + 1, v) \end{aligned}$$

# Image Coordinates System

**Linear Interpolation**

- one-dimensional
- two-dimensional

$$I(u, v) = (a_1 u + a_0)(b_1 v + b_0)$$



# Image Coordinates System

**Linear Interpolation**

- one-dimensional
- two-dimensional

$$I(u,v) = (a_1 u + a_0)(b_1 v + b_0)$$

$$\Updownarrow$$

$$\begin{aligned} I(u,v) &= [(\lfloor u \rfloor + 1 - u)k_1 + (u - \lfloor u \rfloor)k_2][(\lfloor v \rfloor + 1 - v)k_3 + (v - \lfloor v \rfloor)k_4] \\ &= [(1-\Delta u)k_1 + \Delta u k_2][(1-\Delta v)k_3 + \Delta v k_4], \end{aligned}$$

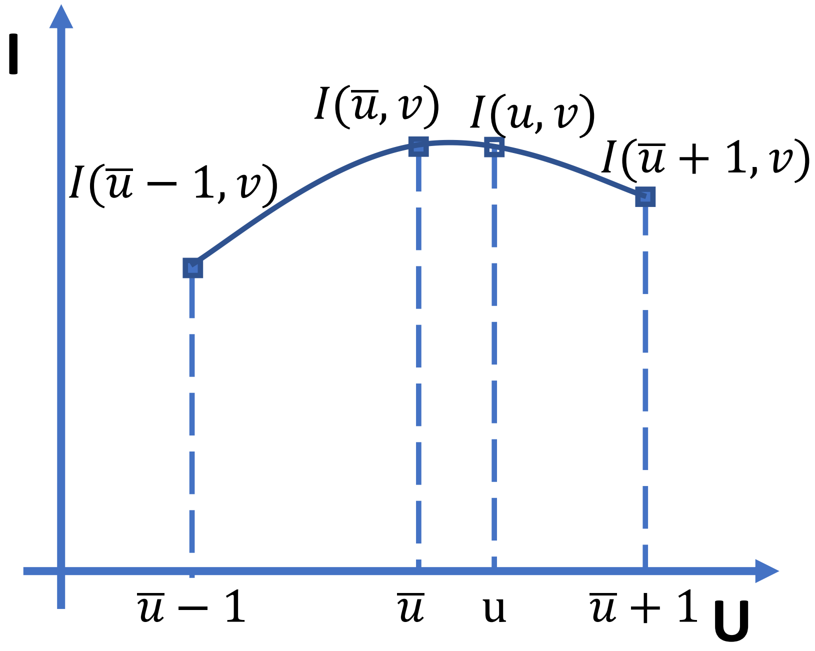
$$\begin{cases} ((1-\Delta u)k_1 + \Delta u k_2)((1-\Delta v)k_3 + \Delta v k_4)|_{u=\lfloor u \rfloor, v=\lfloor v \rfloor} & = I(\lfloor u \rfloor, \lfloor v \rfloor) \\ ((1-\Delta u)k_1 + \Delta u k_2)((1-\Delta v)k_3 + \Delta v k_4)|_{u=\lfloor u \rfloor+1, v=\lfloor v \rfloor} & = I(\lfloor u \rfloor + 1, \lfloor v \rfloor) \\ ((1-\Delta u)k_1 + \Delta u k_2)((1-\Delta v)k_3 + \Delta v k_4)|_{u=\lfloor u \rfloor, v=\lfloor v \rfloor+1} & = I(\lfloor u \rfloor, \lfloor v \rfloor + 1) \\ ((1-\Delta u)k_1 + \Delta u k_2)((1-\Delta v)k_3 + \Delta v k_4)|_{u=\lfloor u \rfloor+1, v=\lfloor v \rfloor+1} & = I(\lfloor u \rfloor + 1, \lfloor v \rfloor + 1) \end{cases} \Rightarrow \begin{aligned} k_1 k_3 &= I(\lfloor u \rfloor, \lfloor v \rfloor), \\ k_2 k_3 &= I(\lfloor u \rfloor + 1, \lfloor v \rfloor), \\ k_1 k_4 &= I(\lfloor u \rfloor, \lfloor v \rfloor + 1), \\ k_2 k_4 &= I(\lfloor u \rfloor + 1, \lfloor v \rfloor + 1). \end{aligned}$$

$$\begin{aligned} &= (\lfloor u \rfloor + 1 - u)(\lfloor v \rfloor + 1 - v)I(\lfloor u \rfloor, \lfloor v \rfloor) + (u - \lfloor u \rfloor)(\lfloor v \rfloor + 1 - v)I(\lfloor u \rfloor + 1, \lfloor v \rfloor) \\ &\quad + (\lfloor u \rfloor + 1 - u)(v - \lfloor v \rfloor)I(\lfloor u \rfloor, \lfloor v \rfloor + 1) + (u - \lfloor u \rfloor)(v - \lfloor v \rfloor)I(\lfloor u \rfloor + 1, \lfloor v \rfloor + 1) \end{aligned}$$

# Image Coordinates System

**Quadratic Interpolation**

- one-dimensional
- two-dimensional




$$I(u,v) = a_2 u^2 + a_1 u + a_0$$

$$\Updownarrow$$

$$I(u,v) = (u-\bar{u})^2 k_2 + (u-\bar{u})k_1 + k_0$$

"bar" on head denotes the round function

$$\begin{cases} (u-\bar{u})^2 k_2 + (u-\bar{u})k_1 + k_0|_{u=\bar{u}-1} & = I(\bar{u}-1, v) \\ (u-\bar{u})^2 k_2 + (u-\bar{u})k_1 + k_0|_{u=\bar{u}} & = I(\bar{u}, v) \\ (u-\bar{u})^2 k_2 + (u-\bar{u})k_1 + k_0|_{u=\bar{u}+1} & = I(\bar{u}+1, v) \end{cases}$$

$$\begin{cases} k_2 - k_1 + k_0 & = I(\bar{u}-1, v) \\ k_0 & = I(\bar{u}, v) \\ k_2 + k_1 + k_0 & = I(\bar{u}+1, v) \end{cases}$$

$$k_2 = \frac{I(\bar{u}+1, v) + I(\bar{u}-1, v) - 2I(\bar{u}, v)}{2}$$

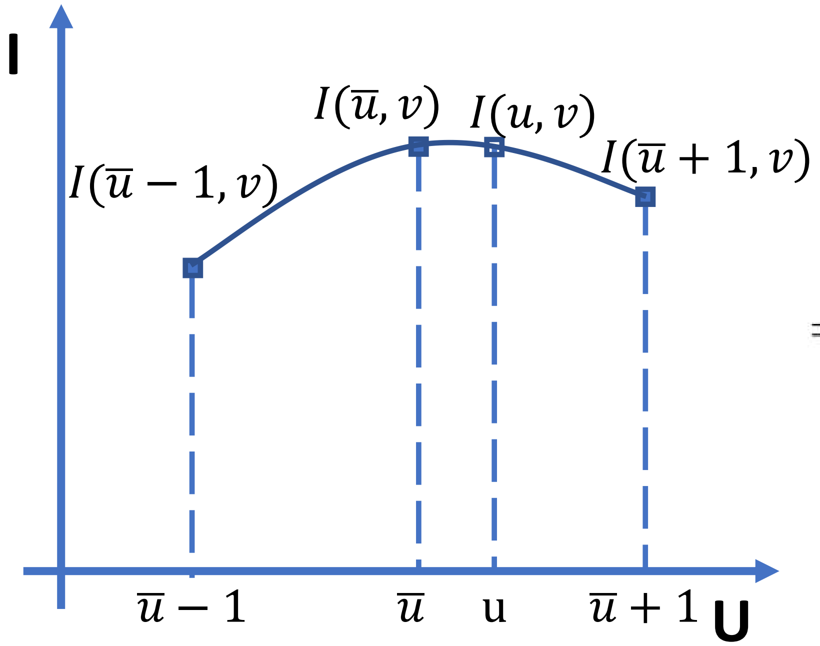
$$k_1 = \frac{I(\bar{u}+1, v) - I(\bar{u}-1, v)}{2}$$

$$k_0 = I(\bar{u}, v).$$

# Image Coordinates System

**Quadratic Interpolation**

- one-dimensional
- two-dimensional




$$I(u,v) = a_2u^2 + a_1u + a_0$$

$$\Updownarrow$$

$$I(u,v) = (u-\bar{u})^2k_2 + (u-\bar{u})k_1 + k_0$$

"bar" on head denotes the round function

$$= (u-\bar{u})^2\frac{I(\bar{u}+1,v)+I(\bar{u}-1,v)-2I(\bar{u},v)}{2} + (u-\bar{u})\frac{I(\bar{u}+1,v)-I(\bar{u}-1,v)}{2} + I(\bar{u},v)$$

# Image Coordinates System

**Quadratic Interpolation**

- one-dimensional
- two-dimensional

$$I(u,v) = (a_2u^2 + a_1u + a_0)(b_2v^2 + b_1v + b_0)$$

$$\Updownarrow$$

$$\begin{aligned} I(u,v) &= [(u-\bar{u})^2k_2 + (u-\bar{u})k_1 + k_0][(v-\bar{v})^2k_5 + (v-\bar{v})k_4 + k_3] \\ &= (\Delta u^2k_2 + \Delta uk_1 + k_0)(\Delta v^2k_5 + \Delta vk_4 + k_3) \\ &= \Delta u^2\Delta v^2k_2k_5 + \Delta u^2\Delta vk_2k_4 + \Delta u^2k_2k_3 \\ &\quad + \Delta u\Delta v^2k_1k_5 + \Delta u\Delta vk_1k_4 + \Delta uk_1k_3 + \Delta v^2k_0k_5 + \Delta vk_0k_4 + k_0k_3 \end{aligned}$$

$$\begin{aligned} &\Delta u^2\Delta v^2k_2k_5 + \Delta u^2\Delta vk_2k_4 + \Delta u^2k_2k_3 + \Delta u\Delta v^2k_1k_5 + \Delta u\Delta vk_1k_4 + \Delta uk_1k_3 \\ &\quad + \Delta v^2k_0k_5 + \Delta vk_0k_4 + k_0k_3|_{u=\bar{u}+i,v=\bar{v}+j} = I(\bar{u}+i,\bar{v}+j) \end{aligned}$$

# Image Coordinates System

**Quadratic Interpolation**

$$I(u,v) = (a_2u^2 + a_1u + a_0)(b_2v^2 + b_1v + b_0)$$

- one-dimensional
- two-dimensional

$$\begin{aligned}&\Delta u^2\Delta v^2 k_2k_5 + \Delta u^2\Delta v k_2k_4 + \Delta u^2 k_2k_3 + \Delta u\Delta v^2 k_1k_5 + \Delta u\Delta v k_1k_4 + \Delta u k_1k_3 \\ &\quad + \Delta v^2 k_0k_5 + \Delta v k_0k_4 + k_0k_3|_{u=\bar{u}+i, v=\bar{v}+j} = I(\bar{u}+i, \bar{v}+j)\end{aligned}$$

$$\begin{bmatrix} 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 & 1 \\ 1 & -1 & 1 & 1 & -1 & 1 & 1 & -1 & 1 \\ 0 & 0 & 1 & 0 & 0 & -1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & -1 & -1 & -1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} k_2k_5 \\ k_2k_4 \\ k_2k_3 \\ k_1k_5 \\ k_1k_4 \\ k_1k_3 \\ k_0k_5 \\ k_0k_4 \\ k_0k_3 \end{bmatrix} = \begin{bmatrix} I(\bar{u}-1, \bar{v}-1) \\ I(\bar{u}, \bar{v}-1) \\ I(\bar{u}+1, \bar{v}-1) \\ I(\bar{u}-1, \bar{v}) \\ I(\bar{u}, \bar{v}) \\ I(\bar{u}+1, \bar{v}) \\ I(\bar{u}-1, \bar{v}+1) \\ I(\bar{u}, \bar{v}+1) \\ I(\bar{u}+1, \bar{v}+1) \end{bmatrix}$$

# Image Coordinates System

**Quadratic Interpolation**

$$I(u,v) = (a_2u^2 + a_1u + a_0)(b_2v^2 + b_1v + b_0)$$

- one-dimensional
- two-dimensional

$$\begin{aligned}&\Delta u^2\Delta v^2 k_2k_5 + \Delta u^2\Delta v k_2k_4 + \Delta u^2 k_2k_3 + \Delta u\Delta v^2 k_1k_5 + \Delta u\Delta v k_1k_4 + \Delta u k_1k_3 \\ &\quad + \Delta v^2 k_0k_5 + \Delta v k_0k_4 + k_0k_3|_{u=\bar{u}+i,v=\bar{v}+j} = I(\bar{u}+i,\bar{v}+j)\end{aligned}$$

$$\mathbf{K} = \mathbf{C}^{-1}\mathbf{I}_{\bar{u},\bar{v}}$$

$$\begin{bmatrix} k_2k_5 \\ k_2k_4 \\ k_2k_3 \\ k_1k_5 \\ k_1k_4 \\ k_1k_3 \\ k_0k_5 \\ k_0k_4 \\ k_0k_3 \end{bmatrix} = \begin{bmatrix} \frac{1}{4} & -\frac{1}{2} & \frac{1}{4} & -\frac{1}{2} & 1 & -\frac{1}{2} & \frac{1}{4} & -\frac{1}{2} & \frac{1}{4} \\ -\frac{1}{4} & \frac{1}{2} & -\frac{1}{4} & 0 & 0 & 0 & \frac{1}{4} & -\frac{1}{2} & \frac{1}{4} \\ 0 & 0 & 0 & \frac{1}{2} & -1 & \frac{1}{2} & 0 & 0 & 0 \\ -\frac{1}{4} & 0 & \frac{1}{4} & \frac{1}{2} & 0 & -\frac{1}{2} & -\frac{1}{4} & 0 & \frac{1}{4} \\ \frac{1}{4} & 0 & -\frac{1}{4} & 0 & 0 & 0 & -\frac{1}{4} & 0 & \frac{1}{4} \\ 0 & 0 & 0 & -\frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & -1 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} I(\bar{u}-1,\bar{v}-1) \\ I(\bar{u},\bar{v}-1) \\ I(\bar{u}+1,\bar{v}-1) \\ I(\bar{u}-1,\bar{v}) \\ I(\bar{u},\bar{v}) \\ I(\bar{u}+1,\bar{v}) \\ I(\bar{u}-1,\bar{v}+1) \\ I(\bar{u},\bar{v}+1) \\ I(\bar{u}+1,\bar{v}+1) \end{bmatrix}$$

# Image Coordinates System

**Quadratic Interpolation**

$$I(u,v) = (a_2u^2 + a_1u + a_0)(b_2v^2 + b_1v + b_0)$$

- one-dimensional
- two-dimensional

$$\begin{aligned} I(u,v) &= \left(\begin{bmatrix}\Delta u^2 & \Delta u & 1\end{bmatrix} \otimes \begin{bmatrix}\Delta v^2 & \Delta v & 1\end{bmatrix}\right)\mathbf{C}^{-1}\mathbf{I}_{\bar{u},\bar{v}} \\ &= \left(\begin{bmatrix}\Delta u^2 & \Delta u & 1\end{bmatrix} \otimes \begin{bmatrix}\Delta v^2 & \Delta v & 1\end{bmatrix}\right) \\ &\quad \begin{bmatrix} \frac{1}{4} & -\frac{1}{2} & \frac{1}{4} & -\frac{1}{2} & 1 & -\frac{1}{2} & \frac{1}{4} & -\frac{1}{2} & \frac{1}{4} \\ -\frac{1}{4} & \frac{1}{2} & -\frac{1}{4} & 0 & 0 & 0 & \frac{1}{4} & -\frac{1}{2} & \frac{1}{4} \\ 0 & 0 & 0 & \frac{1}{2} & -1 & \frac{1}{2} & 0 & 0 & 0 \\ -\frac{1}{4} & 0 & \frac{1}{4} & \frac{1}{2} & 0 & -\frac{1}{2} & -\frac{1}{4} & 0 & \frac{1}{4} \\ \frac{1}{4} & 0 & -\frac{1}{4} & 0 & 0 & 0 & -\frac{1}{4} & 0 & \frac{1}{4} \\ 0 & 0 & 0 & -\frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & -1 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} I(\bar{u}-1,\bar{v}-1) \\ I(\bar{u},\bar{v}-1) \\ I(\bar{u}+1,\bar{v}-1) \\ I(\bar{u}-1,\bar{v}) \\ I(\bar{u},\bar{v}) \\ I(\bar{u}+1,\bar{v}) \\ I(\bar{u}-1,\bar{v}+1) \\ I(\bar{u},\bar{v}+1) \\ I(\bar{u}+1,\bar{v}+1) \end{bmatrix} \end{aligned}$$

$\Delta u \equiv u - \bar{u}$, and $\Delta v \equiv v - \bar{v}$.

## Image Coordinates System

**Quadratic Interpolation**

- image resolution enhancement

Download *NonMaximumSuppression_practice.m* and complete it

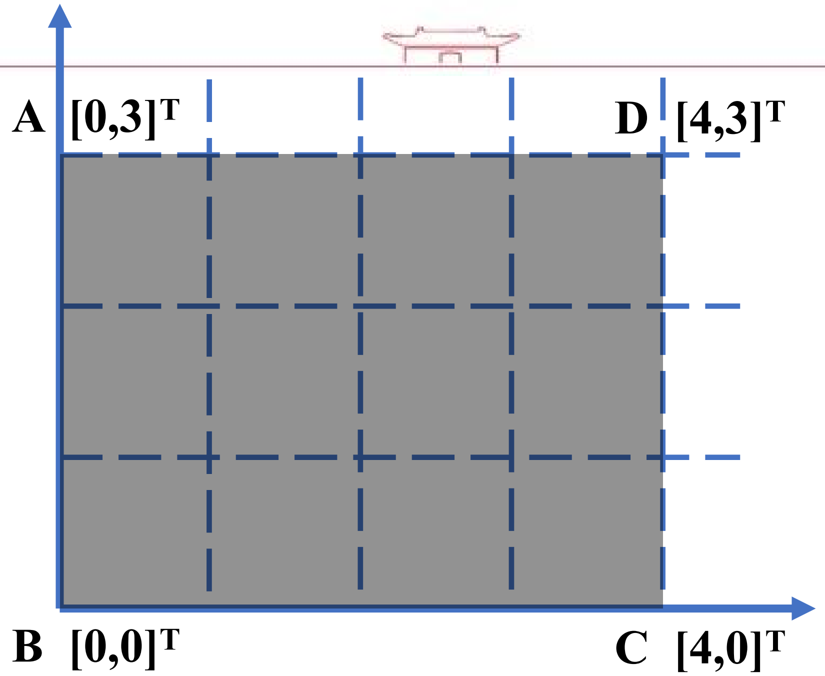
Download *demoBiquadraticInterpolation.m* and run it

DISCUSSION

Q & A

# Comprehensive Practice

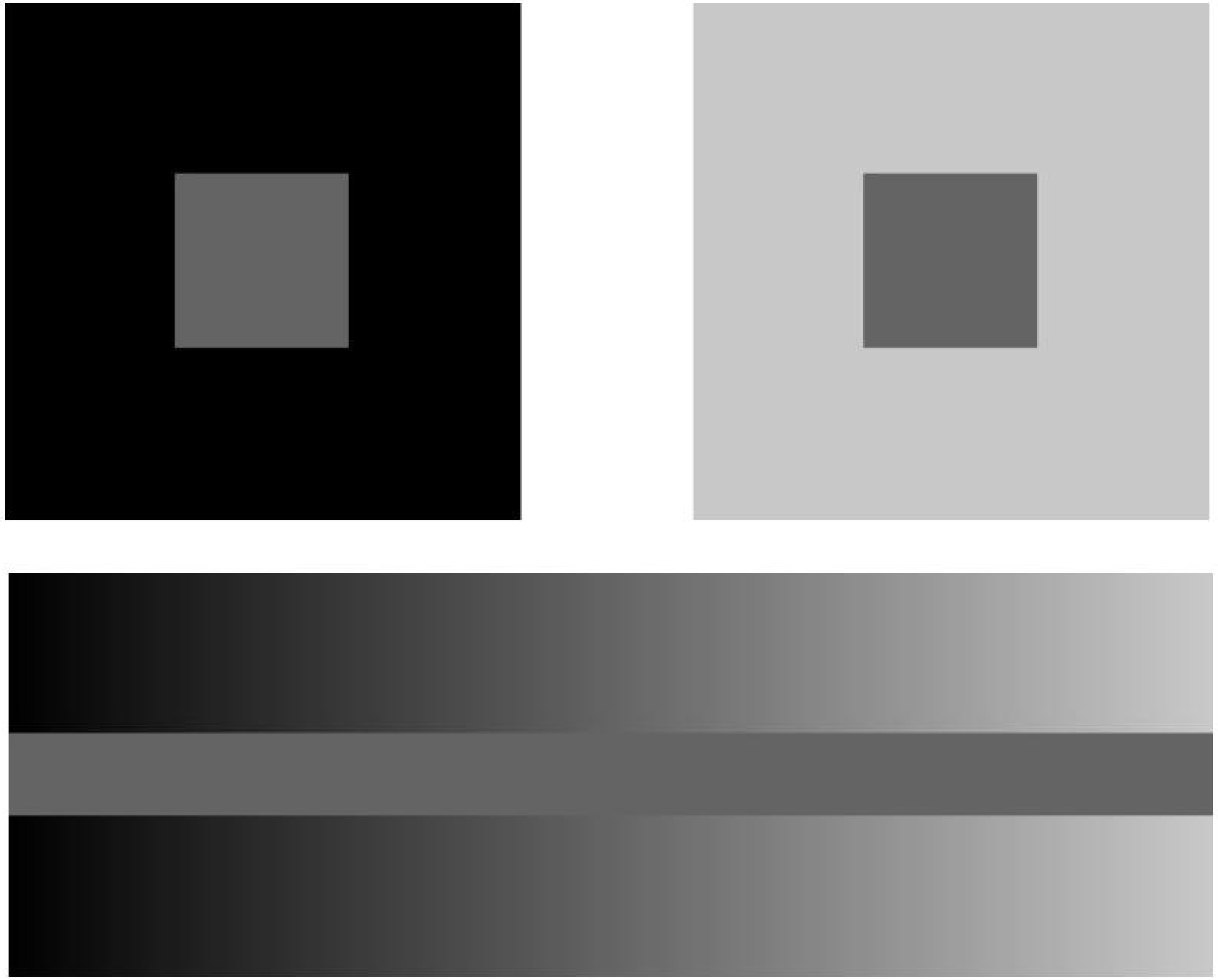
- Adjustable Rectangle Generation
  - Given a rectangle with its shape characterized by the illustrated corner points A, B, C, and D
  - For visualization of the rectangle actually in a GUI
    - Generate a white image of 300 x 300 pixels
    - The rectangle center should be located at the pixel (180,150) namely the 150-th row and 180-th column pixel
    - Suppose the visualization scaling factor is 57.7
    - For pixels internal to the rectangle, their pixel value should be 0 namely black
    - For pixels external to the rectangle, their pixel value should be 255 namely white
    - For a boundary pixel (partially internal and partially external to the rectangle), its pixel value should be a weighted average of both color values according to the areas of its black & white parts: {pixel value} = {black part area ratio} x {black value} + {white part area ratio} x {white value}
  - Set "white", "black" to different values (say 200 for white, 50 for black) and re-try



DISCUSSION

Q & A

# Strenghten edge contrasts



# Edge Detection

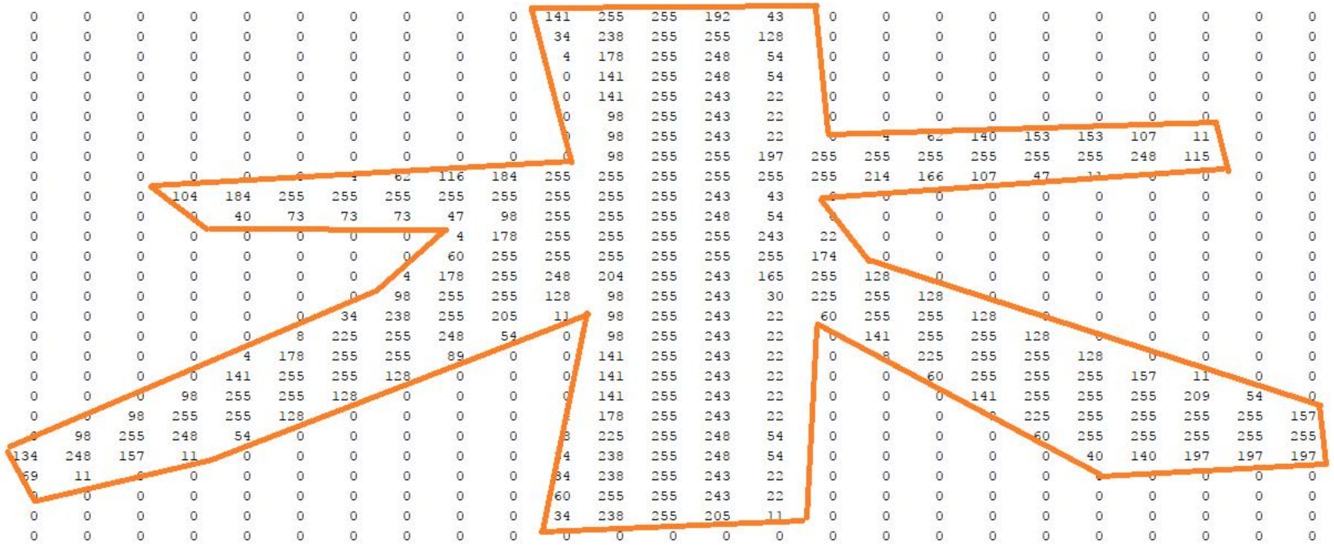
**Gradient Image**

- Horizontal
- Vertical

$$\nabla \bar{I}(u,v) \approx \begin{bmatrix} \frac{I(u+\Delta u,v)-I(u,v)}{\Delta u} \\ \frac{I(u,v+\Delta v)-I(u,v)}{\Delta v} \end{bmatrix}, \quad \begin{bmatrix} u \\ v \end{bmatrix} \in \mathbf{I}_D.$$

$$\nabla I(u,v) = \begin{bmatrix} I_u(u,v) \\ I_v(u,v) \end{bmatrix} \equiv \begin{bmatrix} \frac{\partial I(u,v)}{\partial u} \\ \frac{\partial I(u,v)}{\partial v} \end{bmatrix}$$

$$\nabla \bar{I}(u,v) \approx \begin{bmatrix} \frac{I(u+\Delta u,v)-I(u-\Delta u,v)}{2\Delta u} \\ \frac{I(u,v+\Delta v)-I(u,v-\Delta v)}{2\Delta v} \end{bmatrix}, \quad \begin{bmatrix} u \\ v \end{bmatrix} \in \mathbf{I}_D.$$



# Edge Detection

**Gradient Image**

- Horizontal
- Vertical

$$\nabla I(u,v) = \begin{bmatrix} I_u(u,v) \\ I_v(u,v) \end{bmatrix}$$

$$I(u+\Delta u) = I(u) + I'(u)\Delta u + \frac{I''(u)}{2}\Delta u^2 + o(\Delta u^3)$$

$$I(u-\Delta u) = I(u) - I'(u)\Delta u + \frac{I''(u)}{2}\Delta u^2 + o(\Delta u^3)$$

$$\Delta u = \Delta v = 1$$

$$\nabla \bar{I}(u,v) \approx \begin{bmatrix} \frac{I(u+\Delta u,v)-I(u,v)}{\Delta u} \\ \frac{I(u,v+\Delta v)-I(u,v)}{\Delta v} \end{bmatrix}, \quad \begin{bmatrix} u \\ v \end{bmatrix} \in \mathbf{I}_D.$$

$$\frac{I(u+\Delta u) - I(u)}{\Delta u} = I'(u) + \frac{I''(u)}{2}\Delta u + o(\Delta u^2) = I'(u) + o(\Delta u)$$

$$\nabla \bar{I}(u,v) \approx \begin{bmatrix} \frac{I(u+\Delta u,v)-I(u-\Delta u,v)}{2\Delta u} \\ \frac{I(u,v+\Delta v)-I(u,v-\Delta v)}{2\Delta v} \end{bmatrix}, \quad \begin{bmatrix} u \\ v \end{bmatrix} \in \mathbf{I}_D.$$

$$\frac{I(u+\Delta u) - I(u-\Delta u)}{2\Delta u} = I'(u) + o(\Delta u^2)$$

$$\bar{I}_u(u,v) \approx \begin{bmatrix} \frac{1}{2} & 0 & -\frac{1}{2} \end{bmatrix} * \bar{I}(u,v), \quad \bar{I}_v(u,v) \approx \begin{bmatrix} -\frac{1}{2} \\ 0 \\ \frac{1}{2} \end{bmatrix} * \bar{I}(u,v).$$

$$\begin{bmatrix} \frac{1}{2} & 0 & -\frac{1}{2} \end{bmatrix} \equiv \begin{cases} -\frac{1}{2} & u = 1, v = 0 \\ \frac{1}{2} & u = -1, v = 0 \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{bmatrix} -\frac{1}{2} \\ 0 \\ \frac{1}{2} \end{bmatrix} \equiv \begin{cases} -\frac{1}{2} & u = 0, v = 1 \\ \frac{1}{2} & u = 0, v = -1 \\ 0 & \text{otherwise} \end{cases}$$

# Edge Detection

**Gradient Image**

- Horizontal
- Vertical

$$\nabla I(u,v) = \begin{bmatrix} I_u(u,v) \\ I_v(u,v) \end{bmatrix}$$

$$\nabla \bar{I}(u,v) \approx \begin{bmatrix} \frac{I(u+1,v)-I(u-1,v)}{2} \\ \frac{I(u,v+1)-I(u,v-1)}{2} \end{bmatrix}$$

$$\Delta u = \Delta v = 1$$

$$\bar{I}_u(u,v) \approx \begin{bmatrix} \frac{1}{2} & 0 & -\frac{1}{2} \end{bmatrix} * \bar{I}(u,v), \quad \bar{I}_v(u,v) \approx \begin{bmatrix} -\frac{1}{2} \\ 0 \\ \frac{1}{2} \end{bmatrix} * \bar{I}(u,v).$$

$$\nabla \bar{I}(u,v) \approx \begin{bmatrix} \frac{I(u+1,v+1)+2I(u+1,v)+I(u+1,v-1)-I(u-1,v+1)-2I(u-1,v)-I(u-1,v-1)}{8} \\ \frac{I(u+1,v+1)+2I(u,v+1)+I(u-1,v+1)-I(u+1,v-1)-2I(u,v-1)-I(u-1,v-1)}{8} \end{bmatrix}$$

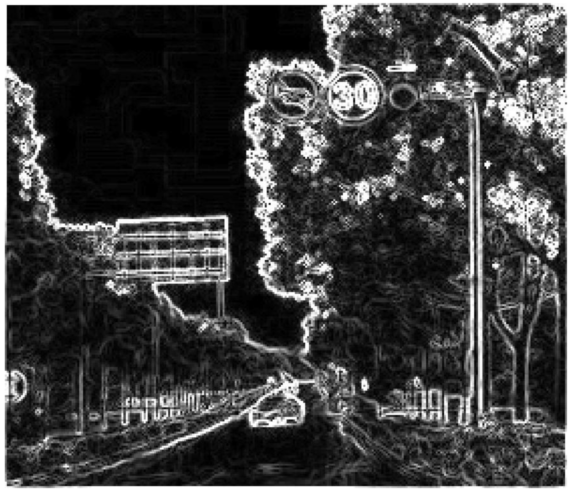
$$\bar{I}_u(u,v) \approx \begin{bmatrix} \frac{1}{8} & 0 & -\frac{1}{8} \\ \frac{1}{4} & 0 & -\frac{1}{4} \\ \frac{1}{8} & 0 & -\frac{1}{8} \end{bmatrix} * \bar{I}(u,v), \quad \bar{I}_v(u,v) \approx \begin{bmatrix} -\frac{1}{8} & -\frac{1}{4} & -\frac{1}{8} \\ 0 & 0 & 0 \\ \frac{1}{8} & \frac{1}{4} & \frac{1}{8} \end{bmatrix} * \bar{I}(u,v)$$

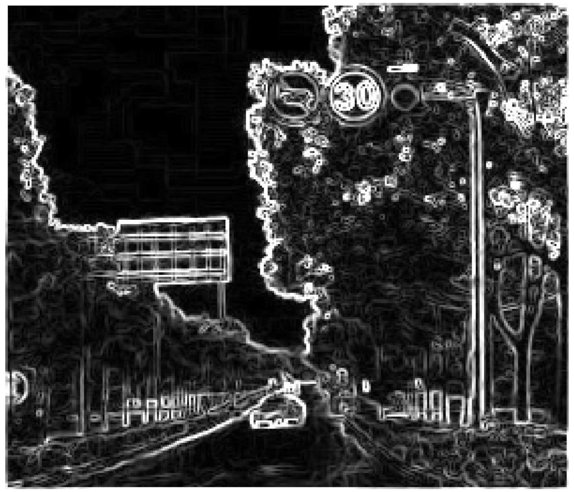
**Sobel operator**

# Edge Detection Practice

**Gradient Image**

- Verify theoretical soundness of the Sobel method
- Try computation of the gradient image
- Can you effectively extract edges based on the gradient image?



Download *RTS02.jpg*, *demoED.m*, *GradientImage.m*, and *GradientImageSobel.m*

Learn from *GradientImage.m*, complete *GradientImageSobel.m*, and run *demoED.m*

# Harris Corner Detection

1.Compute the gradient image

$$\nabla \bar{I}(u,v) \equiv \begin{bmatrix} \bar{I}_u(u,v) \\ \bar{I}_v(u,v) \end{bmatrix}$$

2.Compute three auxiliary images

*pixel-wise operation*

$$\bar{I}_{uu}(u,v) = \bar{I}_u(u,v)^2,$$
$$\bar{I}_{uv}(u,v) = \bar{I}_u(u,v)\bar{I}_v(u,v),$$
$$\bar{I}_{vv}(u,v) = \bar{I}_v(u,v)^2.$$

3.Gaussian filtering over the three auxiliary images

$$\hat{I}_{uu}(u,v) = g(u,v) * \bar{I}_{uu}(u,v),$$
$$\hat{I}_{uv}(u,v) = g(u,v) * \bar{I}_{uv}(u,v),$$
$$\hat{I}_{vv}(u,v) = g(u,v) * \bar{I}_{vv}(u,v),$$

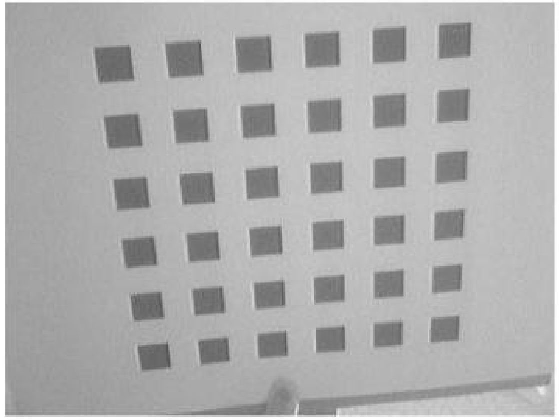
4.Compute the **Harris image**

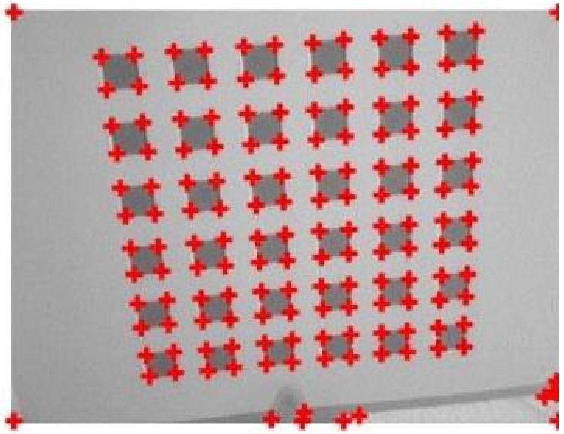
$$\bar{H}(u,v) = \hat{I}_{uu}(u,v)\hat{I}_{vv}(u,v) - \hat{I}_{uv}(u,v)^2 - k[\hat{I}_{uu}(u,v) + \hat{I}_{vv}(u,v)]^2,$$

*k* usually 0.04~0.15 *pixel-wise operation*

5.Apply non-maximum suppression to the Harris image

6.Extract Harris corners via thresholding





DISCUSSION

Q & A